

# Chooladeva Piyasiri

## Data Analyst

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

### PROFILE

Detail-oriented Data Analyst with hands-on experience in data analysis, machine learning, business intelligence, and big data technologies. Skilled in Python, SQL, Power BI, and Tableau, with experience developing predictive models, interactive dashboards, and data-driven solutions. Strong ability to transform complex datasets into actionable insights that support business and operational decision-making.

### AREAS OF EXPERTISE

- ANALYTICS & BI:** SQL (SQL Server, MS SSIS), Power BI (DAX, Power Query), Tableau, Excel (Pivot Tables, Dashboards, Lookups)
- PROGRAMMING:** Python (Pandas, NumPy, Scikit-Learn, Seaborn, Matplotlib), R (Tidyverse)
- ML & NLP:** Scikit-Learn, NLTK, spaCy, Hugging Face, PyTorch, OpenCV
- BIG DATA:** Apache Spark, Hadoop (HDFS, Hive), MapReduce

### PROFESSIONAL EXPERIENCE

#### Data Science & AI Intern | June 2024 –December 2024

Scicom Lanka (Pvt.) Ltd – Colombo, Sri Lanka.

- Built an end-to-end automated passenger risk-screening pipeline processing 64,000+ immigration records per flight, replacing manual analyst rule-writing with an ML-driven flagging system.
- Engineered 18+ behavioral and document-based risk features (travel frequency, visa validity, nationality mismatch, last-minute booking patterns) and applied K-Means clustering, Isolation Forest, and an Autoencoder to identify anomalous passengers.
- Generated human-readable flagging rules using FP-Growth association rule mining, enabling non-technical officers to understand and act on each flag with clear plain-English explanations.
- Implemented a PPO reinforcement learning environment to prioritize the most effective flagging rules, laying the groundwork for a self-improving rule-selection system.

#### Data Science Intern | January 2023 –April 2023

Data Glacier Academics & IT Consulting

- Applied Python-based data analysis and machine learning to build, deploy, and govern analytical models, delivering visual insights and effectively communicating results to stakeholders.

### EDUCATION

#### MSc in Big Data Analytics

Robert Gordon University, United Kingdom.

May 2025- Present

- Advanced Databases, Big Data Programming, Cloud Computing, Data Analysis, Data Mining, Research Methods, Data Warehousing, Text Analytics

## BSc (Hons) in Data Science

Coventry University, United Kingdom.

Graduated: August 2024

- Big Data Analytics, Advanced Machine Learning, Data Visualization Using R & Tableau, Data Visualization & Interactive Dashboards Using Python, Data Visualization Using Power BI, SQL Databases and Business Analytics

## SELECTED PROJECTS

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### Sales, Customer, Inventory & Staff Analysis

SQL, SSIS, Power BI

[GitHub Repository](#) | [Power BI Dashboard](#)

- Sales, inventory, and staff data were split across **10+ relational tables**, making reporting slow and inconsistent. Extracted and integrated this data using SQL **JOINS** and **Views** to create a unified reporting dataset.
- Transformed, cleaned, and aggregated data using **SQL** and **SSIS**, improving reporting accuracy and efficiency.
- Developed interactive **Power BI** dashboard that identified sales, inventory, customer, and staff performance trends to support business decision-making.

### Healthcare Performance Analysis Dashboard

Power BI, DAX, Excel

[Power BI Dashboard](#)

- Cleaned and validated **55,500+ patient records** using **Excel** and **DAX** to improve data quality and consistency.
- Designed an interactive **Power BI** dashboard comparing hospitals and insurers across cost, patient volume, and clinical outcome metrics.

### Supermarket Sales Forecasting Pipeline

Python, Scikit-learn, Pandas, Seaborn

[GitHub Repository](#)

- Outlets lacked a reliable way to anticipate demand for high-turnover product categories, leading to under- or over-stocking. Built end-to-end data engineering and machine learning pipelines to forecast next-week sales for these categories.
- Performed **EDA** and engineered predictive features using a modular Python pipeline for scalable processing.
- Developed forecasting model that generated actionable sales predictions to support inventory planning.

### Big Data Weather Analysis & Prediction

Apache Spark, Hive, Hadoop MapReduce, Tableau

[GitHub Repository](#) | [Tableau Dashboard](#)

- Processed weather data (2010–2024) using **Hadoop**, **Hive**, and **Spark** to analyze district-level climate trends and extreme weather events.
- Developed a **Spark MLlib** prediction model to forecast periods of low May evapotranspiration.
- Created a **Tableau** dashboard to communicate climate insights for data-driven planning and decision-making.

### Hotel Review Sentiment Analysis

Python, NLP, BERT, XGBoost

[GitHub Repository](#)

- Collected and analyzed **5,320 TripAdvisor reviews** from **116 hotels** using NLP and machine learning techniques.
- Built and evaluated sentiment classification models using **BERT**, **SVM**, **XGBoost**, **TF-IDF**, and **word embeddings**.
- Applied **BERTopic** for aspect-based sentiment analysis to identify customer opinions on rooms, food, and staff, producing actionable recommendations for hotel management.